Interview Preparation Checklist

Sanskar Jain · Backend Software Engineer · C#/.NET · AWS · Microservices

Programming Languages

C#

- ☐ OOP Principles (Encapsulation, Inheritance, Polymorphism, Abstraction)
- ☐ Value Types vs Reference Types, Boxing/Unboxing
- ☐ Classes, Structs, Records, Interfaces, Abstract Classes
- ☐ Generics and Constraints
- ☐ Delegates, Events, Lambdas, Func/Action
- ☐ LINQ (Deferred Execution, IEnumerable vs IQueryable)
- ☐ Collections (List, Dictionary, HashSet, Concurrent Collections)
- ☐ async/await, Task, ValueTask, Task Parallel Library
- ☐ SynchronizationContext, ConfigureAwait, Deadlocks in Async Code
- ☐ Thread Safety, Locks, Semaphores, Interlocked
- ☐ Garbage Collection, Generations, IDisposable, using Statement
- ☐ Exception Handling Best Practices
- ☐ Nullable Reference Types
- ☐ Extension Methods
- ☐ Modern C# Features (Pattern Matching, Records, Init-Only Properties)

Java

- ☐ OOP in Java
- ☐ JVM Architecture, Heap vs Stack, Garbage Collection
- ☐ Collections Framework (HashMap Internals, ArrayList, LinkedList, TreeMap)
- ☐ equals() and hashCode() Contract
- ☐ Streams API and Functional Interfaces
- ☐ Exceptions (Checked vs Unchecked)
- ☐ Multithreading, ExecutorService, CompletableFuture
- ☐ ThreadLocal
- ☐ Immutability and String Pool
- ☐ Java 17 Features (Records, Sealed Classes, Switch Expressions)

JavaScript & TypeScript

- ☐ Event Loop, Call Stack, Microtasks vs Macrotasks
- ☐ Closures, Scope, Hoisting
- ☐ this Binding
- ☐ Promises and async/await
- ☐ ES6+ Features
- ☐ TypeScript Types, Interfaces, Generics, Union and Utility Types

Python

- ☐ Python Fundamentals and Data Types
- ☐ Lists, Dictionaries, Sets, Comprehensions

SQL

- ☐ Joins, Subqueries, CTEs
- ☐ GROUP BY, HAVING, Aggregations
- ☐ Window Functions
- ☐ Common Query Patterns (Nth Highest, Duplicates, Running Totals)

Data Structures & Algorithms

- ☐ Time and Space Complexity (Big-O)
- ☐ Arrays and Strings
- ☐ Hashing and Hash Maps
- ☐ Two Pointers
- ☐ Sliding Window
- ☐ Prefix Sums
- ☐ Binary Search
- ☐ Sorting Algorithms
- ☐ Linked Lists
- ☐ Stacks and Queues, Monotonic Stack
- ☐ Recursion and Backtracking
- ☐ Trees, Binary Trees, Binary Search Trees
- ☐ Tree Traversals (DFS, BFS)
- ☐ Heaps and Priority Queues
- ☐ Tries
- ☐ Graphs (BFS, DFS, Cycle Detection)
- ☐ Topological Sort
- ☐ Shortest Path (Dijkstra)
- ☐ Union-Find
- ☐ Greedy Algorithms
- ☐ Intervals
- ☐ Dynamic Programming
- ☐ Bit Manipulation
- ☐ LRU Cache Implementation

Computer Science Fundamentals

Operating Systems

- ☐ Processes vs Threads
- ☐ Concurrency, Race Conditions, Deadlocks
- ☐ Mutex vs Semaphore
- ☐ Memory Management, Paging, Virtual Memory
- ☐ Context Switching

Computer Networks

- ☐ OSI and TCP/IP Models
- ☐ TCP vs UDP
- ☐ HTTP/1.1 vs HTTP/2, HTTPS and TLS Handshake
- ☐ DNS Resolution
- ☐ HTTP Methods, Headers, Status Codes
- ☐ Cookies, Sessions
- ☐ CORS
- ☐ What Happens When You Type a URL

Software Design Principles

- ☐ SOLID Principles
- ☐ DRY, KISS, YAGNI
- ☐ Dependency Injection and Inversion of Control
- ☐ Composition vs Inheritance
- ☐ Clean Architecture and Layered Architecture

Design Patterns

- ☐ Repository Pattern
- ☐ Factory Pattern
- ☐ Singleton Pattern
- ☐ Strategy Pattern
- ☐ Decorator Pattern
- ☐ Observer Pattern
- ☐ Builder Pattern
- ☐ Adapter Pattern
- ☐ Unit of Work

Backend Development

.NET & ASP.NET Core

- ☐ .NET 8 vs .NET Framework
- ☐ ASP.NET Core Request Pipeline and Middleware
- ☐ Dependency Injection Lifetimes (Singleton, Scoped, Transient)
- ☐ Routing (Attribute vs Conventional)
- ☐ Controllers vs Minimal APIs

- ☐ Model Binding and Model Validation
- ☐ Filters (Action, Exception, Authorization)
- ☐ Global Exception Handling and ProblemDetails
- ☐ Configuration and Options Pattern
- ☐ Logging in ASP.NET Core
- ☐ Kestrel and Hosting Model
- ☐ Background Services and IHostedService
- ☐ HttpClientFactory
- ☐ Authentication and Authorization Middleware
- ☐ JWT Bearer Authentication in ASP.NET Core
- ☐ Forms Authentication (Legacy)
- ☐ ASP.NET MVC and Web API 2
- ☐ Migrating .NET Framework to .NET 8
- ☐ Async Performance Patterns in ASP.NET Core

Spring Boot

- ☐ Spring IoC Container and Bean Lifecycle
- ☐ Bean Scopes
- ☐ Auto-Configuration and Starters
- ☐ Spring MVC Request Flow
- ☐ Exception Handling (@ControllerAdvice)
- ☐ Spring Security Filter Chain
- ☐ Method Security (@PreAuthorize)
- ☐ Spring Data JPA and Repositories
- ☐ Hibernate Entity Lifecycle and Persistence Context
- ☐ Lazy vs Eager Loading, N+1 Problem
- ☐ Hibernate Filters
- ☐ @Transactional and Transaction Propagation
- ☐ Spring AOP (Aspects, Pointcuts, Advice)
- ☐ @Scheduled Jobs
- ☐ Maven Build Lifecycle

REST API Design

- ☐ REST Principles and Constraints
- ☐ Resource Naming
- ☐ HTTP Method Semantics and Idempotency
- ☐ Status Code Selection (including 409 Conflict)
- ☐ API Versioning Strategies (URL, Header, Query)
- ☐ Backward Compatibility and Deprecation
- ☐ Pagination, Filtering, Sorting
- ☐ Request Validation and Error Response Design
- ☐ Rate Limiting and Throttling
- ☐ Swagger/OpenAPI Documentation
- ☐ Webhook Design and Delivery Contracts
- ☐ Web Services Fundamentals

Microservices & Distributed Systems

- ☐ Monolith vs Microservices
- ☐ Service Boundaries and Domain Decomposition
- ☐ Stateless Services and Horizontal Scaling
- ☐ Synchronous vs Asynchronous Communication
- ☐ API Gateway Pattern
- ☐ Load Balancing
- ☐ Event-Driven Architecture
- ☐ Publish/Subscribe and Fan-Out
- ☐ CQRS
- ☐ CAP Theorem
- ☐ Consistency Models and Eventual Consistency
- ☐ Delivery Semantics (At-Most-Once, At-Least-Once, Exactly-Once)
- ☐ Idempotency and Idempotency Keys
- ☐ Deduplication Strategies
- ☐ Retry with Exponential Backoff and Jitter
- ☐ Dead-Letter Queues and Redrive
- ☐ Fault Tolerance and Graceful Degradation
- ☐ Message Ordering
- ☐ Reconciliation and Polling Patterns
- ☐ Multi-Tenant SaaS Architecture

Messaging Systems

Apache Kafka

- ☐ Topics, Partitions, Offsets
- ☐ Producers and Consumers
- ☐ Consumer Groups and Rebalancing
- ☐ Ordering Guarantees
- ☐ Replication and Durability
- ☐ Kafka vs Traditional Message Queues

AWS SQS & SNS

- ☐ SQS Standard vs FIFO Queues
- ☐ Visibility Timeout
- ☐ Long Polling vs Short Polling
- ☐ SQS Dead-Letter Queues
- ☐ SNS Topics and Subscriptions
- ☐ SNS + SQS Fan-Out Pattern
- ☐ Message Filtering
- ☐ SQS vs SNS vs Kafka

Caching

- ☐ Caching Strategies (Cache-Aside, Write-Through, Write-Behind)
- ☐ Cache Invalidation
- ☐ TTL and Eviction Policies (LRU, LFU)
- ☐ Cache Stampede and Thundering Herd
- ☐ Redis Data Structures
- ☐ Redis Persistence and Replication
- ☐ Redis as a Session Store
- ☐ Amazon ElastiCache
- ☐ Distributed Caching vs In-Memory Caching
- ☐ Connection Pooling

Databases

Relational Databases (PostgreSQL, MySQL)

- ☐ Normalization and Denormalization
- ☐ Schema Design
- ☐ ACID Properties
- ☐ Transaction Isolation Levels
- ☐ Locking and Concurrency Control
- ☐ Indexing (B-Tree, Composite, Covering)
- ☐ Query Execution Plans (EXPLAIN)
- ☐ Slow Query Analysis and Optimization
- ☐ Primary Keys, Foreign Keys, Constraints
- ☐ Replication and Read Replicas
- ☐ PostgreSQL vs MySQL

MongoDB

- ☐ Document Data Modeling (Embedding vs Referencing)
- ☐ Indexes in MongoDB
- ☐ Aggregation Pipeline
- ☐ Replica Sets
- ☐ Sharding
- ☐ MongoDB Transactions

Elasticsearch

- ☐ Inverted Index
- ☐ Mappings and Analyzers
- ☐ Shards and Replicas
- ☐ Full-Text Search and Relevance Scoring
- ☐ Search Query Optimization

Data Management

- ☐ SQL vs NoSQL Trade-offs
- ☐ Database Migrations
- ☐ Zero-Downtime Schema and Data Migrations
- ☐ Bulk Data Migration at Scale
- ☐ Multi-Tenant Data Isolation Patterns (Shared Schema, Schema-per-Tenant, Database-per-Tenant)

- ☐ Supabase Managed PostgreSQL

Security & Identity

- ☐ Authentication vs Authorization
- ☐ OAuth 2.0 Roles and Grant Types
- ☐ Authorization Code Flow with PKCE
- ☐ Client Credentials Flow
- ☐ OpenID Connect (ID Token, UserInfo, Discovery Metadata)
- ☐ OIDC Front-Channel vs Back-Channel Logout
- ☐ Single Sign-On and Single Sign-Out
- ☐ JWT Structure, Signing Algorithms, Validation
- ☐ JWT Claims (including JTI)
- ☐ Access Tokens vs Refresh Tokens
- ☐ Token Expiry, Rotation, Revocation
- ☐ Token Blocklists
- ☐ Stateless vs Stateful Sessions
- ☐ Server-Side Session Management
- ☐ Token Replay Attacks
- ☐ Okta (Tenants, Authorization Servers, Event Hooks, System Log)
- ☐ Identity Server
- ☐ Kong API Gateway (Routing, Plugins, Authentication)
- ☐ Role-Based Access Control
- ☐ API Key Authentication
- ☐ Webhook Security and Verification
- ☐ OWASP Top 10
- ☐ XSS, CSRF, SQL Injection
- ☐ Password Hashing
- ☐ Secure Coding Practices

Cloud (AWS)

- ☐ AWS Global Infrastructure (Regions, Availability Zones)
- ☐ Shared Responsibility Model
- ☐ IAM (Users, Roles, Policies)
- ☐ Amazon S3 (Storage Classes, Presigned URLs, Multipart Upload)
- ☐ AWS Lambda (Triggers, Cold Starts, Concurrency, SQS Batch Processing)
- ☐ Amazon EKS
- ☐ Amazon ECS
- ☐ EKS vs ECS vs Lambda
- ☐ Amazon ElastiCache
- ☐ Amazon CloudWatch (Logs, Metrics, Alarms)
- ☐ VPC Fundamentals
- ☐ AWS Cloud Practitioner Concepts

DevOps & Infrastructure

Containers & Orchestration

- ☐ Docker Images, Layers, Containers
- ☐ Dockerfile Best Practices and Multi-Stage Builds
- ☐ Docker vs Virtual Machines
- ☐ Kubernetes Architecture
- ☐ Pods, Deployments, ReplicaSets, Services
- ☐ Ingress
- ☐ ConfigMaps and Secrets
- ☐ Liveness and Readiness Probes
- ☐ Horizontal Pod Autoscaling

Infrastructure as Code

- ☐ Terraform Providers, Resources, Modules
- ☐ Terraform State Management
- ☐ Terraform Plan/Apply Workflow

CI/CD

- ☐ CI/CD Pipeline Concepts
- ☐ Azure DevOps Pipelines (Stages, Jobs, Gates, Parallel Jobs)
- ☐ Automated Testing in Pipelines
- ☐ Build Optimization
- ☐ Deployment Strategies (Rolling, Blue-Green, Canary)
- ☐ Zero-Downtime Deployments
- ☐ Render Deployment

Version Control & Tooling

- ☐ Git Branching Strategies
- ☐ Merge vs Rebase
- ☐ Resolving Merge Conflicts
- ☐ Linux Commands and CLI

Observability

- ☐ Logging, Metrics, Tracing (Three Pillars)
- ☐ Structured Logging
- ☐ Distributed Tracing
- ☐ OpenTelemetry
- ☐ Splunk
- ☐ Monitoring and Alerting
- ☐ Production Debugging and Incident Response

Testing

- ☐ Testing Pyramid
- ☐ Unit Testing vs Integration Testing
- ☐ xUnit and NUnit
- ☐ Mocking and Test Doubles
- ☐ Code Coverage
- ☐ Test Automation
- ☐ Load and Resilience Testing
- ☐ Test-Driven Development

Frontend Development

- ☐ React 18 Fundamentals and Component Lifecycle
- ☐ React Hooks (useState, useEffect, useMemo, useCallback, useRef, useContext)
- ☐ Virtual DOM and Reconciliation
- ☐ React 18 Concurrent Features
- ☐ State Management
- ☐ Reusable Component Design
- ☐ TanStack React Query (Caching, Invalidation)
- ☐ Axios Interceptors and Token Refresh Handling
- ☐ TypeScript with React
- ☐ Vite
- ☐ TailwindCSS
- ☐ HTML Semantics and CSS Layouts (Flexbox, Grid)
- ☐ SPA Authentication Handling

System Design

Fundamentals

- ☐ Scalability (Vertical vs Horizontal)
- ☐ Availability and Reliability
- ☐ Latency vs Throughput
- ☐ Back-of-the-Envelope Estimation
- ☐ Load Balancers
- ☐ Database Scaling (Replication, Partitioning, Sharding)
- ☐ Message Queues in System Design
- ☐ Rate Limiter Design
- ☐ API Gateway in System Design
- ☐ Asynchronous Processing Design
- ☐ High-Level and Low-Level Design

Resume-Specific Designs

- ☐ Asynchronous Rules/Compliance Engine
- ☐ Large File Import/Export System
- ☐ Event-Driven Processing Pipeline
- ☐ Authentication and SSO System
- ☐ Single Sign-Out and Token Revocation System
- ☐ Webhook Ingestion System
- ☐ Multi-Tenant SaaS Platform

- ☐ Payroll Calculation System
- ☐ Attendance Tracking System
- ☐ Expense Tracking System
- ☐ Workflow Automation / Ticket Routing System

Common Design Problems

- ☐ URL Shortener
- ☐ Notification System
- ☐ Distributed Cache
- ☐ File Storage Service
- ☐ Chat System

AI-Assisted Development

- ☐ LLM API Integration (OpenAI API)
- ☐ Prompt Engineering
- ☐ LLM Workflow Orchestration
- ☐ AI-Assisted Coding Workflows (Claude Code)
- ☐ Reviewing and Validating AI-Generated Code

Projects & Experience Deep Dives

Verisk Analytics

- ☐ RulesRunner: CQRS + SQS Architecture and 3000ms → 500ms Latency Reduction
- ☐ Technical Design Documentation: Service Boundaries, API Contracts, DLQ, SNS Publishing
- ☐ Fault-Tolerant Pipelines: Zero Message Loss Under Failure
- ☐ Xactimate Import/Export: 30+ Data Types, S3 Presigned URLs, MongoDB
- ☐ ASP.NET Core APIs Serving 200K+ Weekly Requests
- ☐ API Versioning Across 16 v2 Endpoints
- ☐ OAuth 2.0 on EKS Behind Kong with Okta JWT and Identity Server
- ☐ Forms Authentication to JWT Migration with Redis Session Store
- ☐ Dual-Layer Revocation (JTI Blocklist + Session Invalidation)
- ☐ Okta Event Hooks Logout Propagation Across 4 Tenants
- ☐ Okta Back-Channel Logout Investigation and Architecture Decision
- ☐ Webhook Receiver: Idempotency, 3-Second Acknowledgement, Reconciliation Poller
- ☐ 34% API Response Time Improvement via Caching and Query Optimization
- ☐ MongoDB/PostgreSQL Indexing and Elasticsearch Search Optimization
- ☐ 1M+ Record Migration Scripts with Zero Downtime
- ☐ Azure DevOps Pipeline Overhaul (50% Deployment, 33% Build Time Reduction)
- ☐ .NET Framework to .NET 8 Migration (55% Tech Debt, 30% Performance)
- ☐ OpenAI-Based Jira Ticket Automation Agent
- ☐ 90% Code Coverage and Code Review Practices

Attendance & Payroll SaaS

- ☐ Shared-Schema Tenant Isolation with org_id Discriminator
- ☐ Hibernate Filters, ThreadLocal Tenant Context, AOP Aspect
- ☐ Defense-in-Depth Against Cross-Tenant Access
- ☐ JWT Access/Refresh Token Design with Server-Side Persistence
- ☐ RBAC Across 3 Roles
- ☐ Payroll Engine: Monthly, Daily, Hourly Models
- ☐ Proration, Overtime, Advances, Carry-Forward Balances
- ☐ Finalized Record Protection (409 Conflict Rule)
- ☐ Scheduled Backfill with External Cron Fallback
- ☐ Docker Deployment on Render with Supabase PostgreSQL
- ☐ Operating a Production System as Sole Engineer for 15+ Clients

Expense Tracker

- ☐ Microservices Decomposition (5+ Services)
- ☐ Kafka-Based Asynchronous Messaging
- ☐ Kong Gateway Routing, Load Balancing, Authentication
- ☐ Kubernetes Orchestration
- ☐ Load and Resilience Testing with 1000+ Transactions

Resume Metrics

- ☐ Measurement Methodology Behind Each Quantified Achievement
- ☐ Baseline vs Outcome for Each Metric
- ☐ Trade-offs and Alternatives Considered for Each Achievement

Behavioral Questions

- ☐ Tell Me About Yourself
- ☐ Walk Through Your Resume
- ☐ STAR Method
- ☐ Most Challenging Project
- ☐ Most Impactful Contribution
- ☐ Ownership and End-to-End Delivery
- ☐ Unblocking a Stalled Technical Decision
- ☐ Disagreement with a Teammate or Lead
- ☐ Handling Conflict
- ☐ Failure and Lessons Learned
- ☐ Production Incident Handling
- ☐ Working Under Ambiguity
- ☐ Learning a New Technology Quickly
- ☐ Prioritization and Deadlines
- ☐ Receiving and Giving Feedback
- ☐ Code Review Experience
- ☐ Writing Technical Documentation
- ☐ Cross-Team Collaboration
- ☐ Working Independently as a Sole Engineer
- ☐ Handling Client Requirements
- ☐ Agile/Scrum Practices
- ☐ Strengths and Weaknesses
- ☐ Why Leave Current Role
- ☐ Why This Company
- ☐ Career Goals